

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		rate is directly proportional to volume of thiosulfate		1		1	1	
	(d)	(i)		rate at 30°C is 12 (1) time = $\frac{1000}{12} = 83 \text{ s}$ (1) ecf possible	2			2	2	
		(ii)		this is true only to some extent here rate doubles when temperature goes from 20°C to 30°C (1) [also when temperature goes from 10°C to 20°C] rate <u>more than</u> doubles when temperature goes from 30°C to 40°C (1)			2	2	1	
		(iii)		particles have more energy (1) neutral answer – particles move faster more collisions exceed activation energy (1) neutral answer – more successful collisions	2			2		
				Question 10 total	7	5	4	16	5	8